

Roll No. :

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B. Tech (CSE) AIML**

Semester: **III**

Name of the Course: **Fundamentals of Artificial Intelligence and Machine Learning**

Course Code: **TCS 364**

Time: **90 Minutes**

Maximum Marks: 50

Note:

- I. Answer all the questions by choosing any one of the sub questions
- II. Each questions carries 10 marks

Explain the importance of Bayesian Belief Network (BBN) in AI. Consider the following BBN. (10 Marks)

CO1

The BBN diagram shows the following structure and tables:

- Difficulty** (parent of **Grade**):

d^0	d^1
0.6	0.4
- Intelligence** (parent of **Grade** and **SAT**):

i^0	i^1
0.7	0.3
- Grade** (parent of **Letter**):

	g^1	g^2	g^3
i^0, d^0	0.3	0.4	0.3
i^0, d^1	0.05	0.25	0.7
i^1, d^0	0.9	0.08	0.02
i^1, d^1	0.5	0.3	0.2
- SAT** (parent of **Letter**):

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8
- Letter**:

	l^0	l^1
g^1	0.1	0.9
g^2	0.4	0.6
g^3	0.99	0.01

Compute the probability of an event $P(-I, D, G, -S, L)$.

OR

Compare and contrast the forward and backward chaining in knowledge representation with suitable examples.

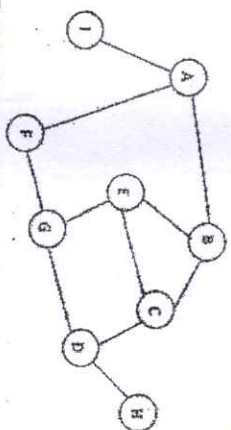
(10 Marks)

Differentiate Artificial Intelligence, Machine Learning, and Deep Learning with suitable examples. Draw a diagram to show their relationship.

CO2

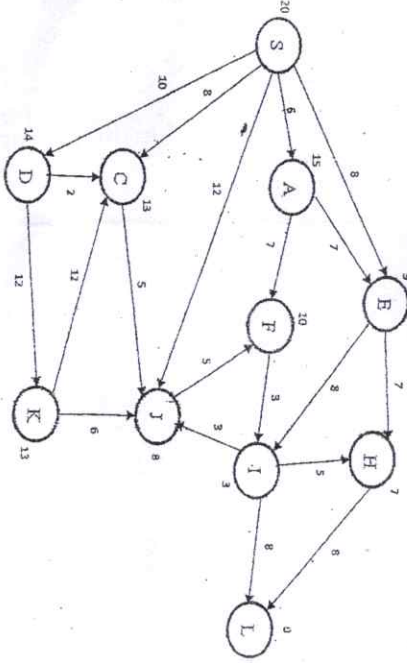
OR

Apply the BFS and DFS algorithms to generate spanning tree for the following graph. Given that A is source.



Q3 (10 Marks)

Discuss the time and space complexity of best first search technique. Apply best first search technique to find the optimal path for the following graph (heuristic). Given that node S is source and L is destination.



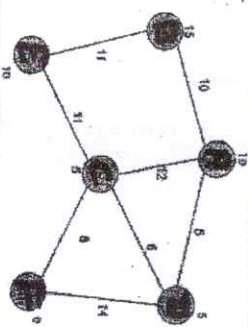
(a)

Discuss the role of AI agents and environments. Consider the example of an autonomous delivery drone and describe:

- (a) The type of agent architecture it uses.
- (b) The characteristics of its environment.

Q4 (10 Marks)

Differentiate A* and AO* algorithm based on their working. Apply the A* algorithm for the following graph, Given that node A is source and F is destination.



(i)

Discuss different types of Learning problems with suitable example. How does each type handle labeled and unlabeled data?

Q5 (10 Marks)

(a) Discuss necessary Exploratory data analysis (EDA) techniques with the help of suitable data/facit/information.

OR

Discuss the importance of First order logic (FOL) in knowledge representation. Compute the following relations from the given table.

- (A) $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$
- (B) $(a \leftrightarrow c) \rightarrow (\neg b \rightarrow (a \wedge c))$
- (C) $(a \wedge b \wedge c) \rightarrow (c \vee a)$
- (D) $a \rightarrow (b \rightarrow a)$

a	b	c
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

(b)